C. Beautiful XOR

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1100 | Tags: bitmasks, constructive algorithms, greedy | https://codeforces.com/problemset/problem/2162/C

You are given two integers a and b. You are allowed to perform the following operation any number of times (including zero):

- choose any integer x such that $0 \le x \le a$ (the current value of a, not initial),
- set $a := a \oplus x$. Here, $\oplus$ represents the bitwise XOR operator.

After performing a sequence of operations, you want the value of a to become exactly b.

Find a sequence of at most 100 operations (values of x used in each operation) that transforms a into b, or report that it is impossible.

Note that you are not required to find the minimum number of operations, but any valid sequence of at most 100 operations.

Input

The first line of input contains a single integer t ($1 \le t \le 1000$) — the number of test cases.

Each test case contains two integers a and b ($1 \le a, b \le 10^9$).

Output

For each test case, if it is impossible to obtain b from a using the allowed operations, print a single line containing -1.

Otherwise, on the first line print a single integer k ($0 \le k \le 100$) — the number of operations. On the second line print k integers $(x_1, x_2, \ldots, x_k)$ — the chosen values of x in the order you apply them.

If there are multiple valid sequences, you may print any one of them.

Example

Input
6 9 6 13 13 292 929 405 400 998 244 244 353

Output

```
2
7 8
0
-1
1
5
2
25 779
-1
```

Note

For the first test case,

- choose $x = 7$, now a becomes equal to $9 \oplus 7 = 14$.
- choose $x = 8$, now a becomes equal to $14 \oplus 8 = 6$.

Thus, we can make $a = b$.

For the fourth test case, choosing $x = 5$ makes $a = b$.